power

J3
UJ20-C-H-G-SMT-TR
R4 5.1K
R1 5.1K
Power Switch
S1
SLW-874744-6A-RA-N-D
POWER_ON
D1 SS24
5V_USB
C16 22uF
U1 USBLC6-2P6
TP1 TestPoint
TP3 TestPoint
TP2 TestPoint
BT1 Battery_Cell
R6 1K
D2 BAT54C
C21 4.7uF
C25 1uF
IC2 TPS73733DCQ
C2 1uF
C6 22uF
C8 22uF
C10 0.01uF
C19 22uF
C20 22uF
C17 1uF
C18 4.7uF
IC4 TPS22918TDBVTQ1
C23 0.001uF

USB PORT Bypass

J2 Conn_01x03
J1 Conn_01x01

TP4056 Connectors

J10 Conn_01x01
J11 Conn_01x01
J12 Conn_01x01
J13 Conn_01x01

Battery Level

R3 33K
R5 33K
C14 0.001uF

Battery Controller Bypass

JP1 SolderJumper_2_Open

Lighting sensor

IC3
AS3935-BQFT
L1 MA5532-AE8
R2 10k
C4 130pF
C7 820pF
C13 10uF
FB1 BLM18AG601SN1D
C11 0.1uF
C15 1uF
R12 100k

STM32

CM7V-T1A-32.768k-7pF-20PPM-TA-QA
Y2
C29 4.7pF
C34 8.2pF
C35 8.2pF
C39 0.1uF
C30 4.7pF
IC5
STM32U585RIT6
C43 0.1uF
C42 4.7uF
C26 0.1uF
C32 0.1uF
C33 0.1uF
C36 0.1uF
C27 4.7uF

Reset/Boot Buttons

S11 TS40-66-33-BK-160-SMT-TR
R22 10k
C53 0.1uF
R21 10k
S6 TS40-66-33-BK-160-SMT-TR
C48 0.1uF

Env sensor

U2
BME280
R17 4.7k
R18 4.7k

Keyboard

S2 TS40-66-33-BK-160-SMT-TR
S3 TS40-66-33-BK-160-SMT-TR
S4 TS40-66-33-BK-160-SMT-TR
S5 TS40-66-33-BK-160-SMT-TR
S7 TS40-66-33-BK-160-SMT-TR
S8 TS40-66-33-BK-160-SMT-TR
S9 TS40-66-33-BK-160-SMT-TR
S10 TS40-66-33-BK-160-SMT-TR
C44 0.1uF
C45 0.1uF
C46 0.1uF
C47 0.1uF
C49 0.1uF
C50 0.1uF
C51 0.1uF
C52 0.1uF

GNNS Module

IC1
MAX-M10M-20B
J4
CONMHF1-SMD-T
C1 0.1uF
C3 10uF
C5 10uF
C9 0.1uF
C12 0.1uF

SD CARD

J5
MEM2075-00-140-01-A
R7 33K
R8 33K
R9 33K
R10 33K
R11 33K
C22 10uF
C24 4.7uF

Screen

R14 100k
IC6
SI2301CDS-T1-BE3
C40 1uF
C41 0.1uF
R15 10k
Q2 2N7002
R20 100k
J17 Conn_01x09
C37 1uF
C38 0.1uF

Connectors

DEBUG

J6 Conn_01x04

SPI

J9 Conn_01x06

GPIO

J15 Conn_01x01
J14 Conn_01x03
J16 Conn_01x01

I2C

J7 Conn_01x04

UART1

J8 Conn_01x03

Other

BZ1 Buzzer
Q1 2N7002
D3 1N4148W
R13 100k
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad

USB Power LED D4

R16 1K
RED

Orange LED D5

R19 680
ORANGE

Buzzer

Sheet: /		
File: trailtrack.kicad_sch		
Title: TrailTrack PCB		
Size: A3	Date: 2026-07-20	**Rev: 1.0**
KiCad E.D.A. 10.0.3		Id: 1/1